

Paper Review Assignment

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E. Arıkan, Channel polarization: A method for constructing capacity achieving codes for symmetric binary input memoryless channels, IEEE Trans. Inform. Theory, Jul. 2009

A review on E. Arikan's "Channel polarisation: A method for constructing capacity achieving codes for symmetric binary input memoryless channels"

Introduction

Channel Polarisation: A method for constructing capacity achieving codes for symmetric binary input memoryless channels by E. Arikan tackles the challenge of constructing capacity-achieving code for reliable communication over noisy channels. It presents a novel method called channel polarisation and a new family of codes called polar codes to address this fundamental problem in information theory.

Summary

The core idea behind this paper is to transform a set of independent copies of a binary-input discrete memoryless channel (D-BMC) into polarised channels. As the number of channel copies grows, these polarised channels become increasingly extreme -- some become nearly noiseless, enabling highly reliable communication, while others become extremely noisy, rendering reliable communication practically impossible. Polar codes are cleverly designed to exploit this polarisation phenomenon, allowing them to approach the theoretical limit of reliable communication, better known as the channel capacity. Arikan also mentioned the Bhattacharyya Parameter as a key tool used in the analysis of channel polarisation as it quantifies the reliability of a channel, i.e. the similarity between the output distributions of a channel for different input symbols. The polarisation process aims to minimise the sum of the Bhattacharyya parameters of the set of synthesised channels.

The channel polarisation process involves 2 main steps -- channel combining and channel splitting. Arikan emphasises on the recursive process of combining and splitting channels to achieve channel polarisation. The process starts with multiple independent copies of the original B-DMC and these copies are combined using a specific transformation to create a larger combined channel. Arikan formally describes channel combining as the process of synthesising a vector channel W_N from N copies of the original channel W using a recursive construction. This process relies on a specific generator matrix G_N , constructed using Kronecker products. This combined channel is then split back into multiple binary-input channels, referred to as synthesised channels.

Formally, the vector channel W_N is then split into N binary-input coordinate channels, denoted as $W_N^{(i)}$ for $i = 1, 2, \dots, N$, representing the polarised channels. A very important theorem, Theorem 1, is then introduced and it states that as the number of channels N grows large, approaching infinity, the fraction of channels that are almost perfect approaches the symmetric capacity of the original channel, while the fraction of almost useless channels approaches 1 minus the original channel's symmetric capacity, polarising the channels into 2 extremes. Theorem 2 then focuses on the speed at which polar codes create "good" bit-channels as the code length increases

Theorem 1: For any B-DMC W , the channels $\{W_N^{(i)}\}$ polarize in the sense that, for any fixed $\delta \in (0, 1)$, as N goes to infinity through powers of two, the fraction of indices $i \in \{1, \dots, N\}$ for which $I(W_N^{(i)}) \in (1 - \delta, 1]$ goes to $I(W)$ and the fraction for which $I(W_N^{(i)}) \in [0, \delta)$ goes to $1 - I(W)$.

Theorem 2: For any B-DMC W with $I(W) > 0$, and any fixed $R < I(W)$, there exists a sequence of sets $\mathcal{A}_N \subset \{1, \dots, N\}$, $N \in \{1, 2, \dots, 2^n, \dots\}$, such that $|\mathcal{A}_N| \geq NR$ and $Z(W_N^{(i)}) \leq O(N^{-5/4})$ for all $i \in \mathcal{A}_N$.

Theorem 1 and Theorem 2. Arikan, E. (2009)

Polar codes were then introduced, explaining that these codes leverage the polarisation phenomenon by selecting a subset of the synthesised channels for data transmission. The selection is based on the reliability of the channels, using a criterion related to the Bhattacharyya parameter. More important theorems were also introduced, which were vigorously proved further down.

The recursive nature of the channel combining and splitting was further emphasised as Arikan used a binary tree structure to visually represent the recursive transformation process. This tree, rooted at the original channel, branches out to represent the synthesised channels at each level of recursion. Proposition 5 establishes a recursive relationship for the Bhattacharyya parameters of the synthesised channels, specifically, it shows how the Bhattacharyya parameters at a given level of the tree is dependent on the parameters at the previous level.

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Proposition 5: Suppose $(W, W) \mapsto (W', W'')$ for some set of binary-input channels. Then

$$Z(W'') = Z(W)^2 \quad (26)$$

$$Z(W') \leq 2Z(W) - Z(W)^2 \quad (27)$$

$$Z(W') \geq Z(W) \geq Z(W''). \quad (28)$$

Equality holds in (27) iff W is a BEC. We have $Z(W') = Z(W'')$ iff $Z(W)$ equals 0 or 1, or equivalently, iff $I(W)$ equals 1 or 0.

Proposition 5, Arikan, E. (2009)

The mathematical proofs for the main theorems on channel polarisation were then given, with the proofs utilising the recursive relationships established in previous sections and probabilistic arguments to demonstrate the convergence of the channels' capacities to 0 or 1. The proof of Theorem 1 establishes the fundamental result of channel polarisation -- as the number of channel copies increase, the fraction of almost perfect and almost useless channels approaches the symmetric capacity of the original channel and 1 minus the original channel's capacity, respectively. The proof of Theorem 2 further strengthens the polarisation result by showing that the convergence of the channel's capacities is exponentially fast in the number of channel copies. It introduces the concept of polarising martingale and utilises martingale convergence theorems to prove this result.

Arikan then examines the special case of symmetric channels, which exhibit certain symmetry properties in their transition probabilities, demonstrating that the channel polarisation process preserves the symmetry of channels. Theorem 4, which is specific to symmetric channels, establishes a strong result on the performance of polar codes under SC decoding and states that the block error probability decays exponentially with the block length, highlighting the effectiveness of polar codes for this class of channels.

Theorem 4: For any symmetric B-DMC W and any fixed $R < I(W)$, consider any sequence of G_N -coset codes $(N, K, \mathcal{A}, u_{\mathcal{A}^c})$ with N increasing to infinity, $K = \lfloor NR \rfloor$, $\mathcal{A}$ chosen in accordance with the polar coding rule for W , and $u_{\mathcal{A}^c}$ fixed arbitrarily. The block error probability under successive cancellation decoding satisfies

$$P_e(N, K, \mathcal{A}, u_{\mathcal{A}^c}) = O\left(N^{-\frac{1}{4}}\right). \quad (16)$$

Theorem 4, Arikan, E. (2009)

Propositions 12 and 13 explore the symmetry properties of the channel combining and splitting operations in relation to symmetric channels as Proposition 12 proves that if the original channel is symmetric, then the combined channel is also symmetric and Proposition 13 extends this result to the split channels, demonstrating that they inherit the symmetry property as well.

Proposition 13: If a B-DMC W is symmetric, then the channels W_N and $W_N^{(i)}$ are also symmetric in the sense that

$$\begin{aligned} W_N(y_1^N | u_1^N) &= W_N(a_1^N G_N \cdot y_1^N | u_1^N \oplus a_1^N), \end{aligned} \quad (58)$$

$$\begin{aligned} W_N^{(i)}(y_1^N, u_1^{i-1} | u_i) &= W_N^{(i)}(a_1^N G_N \cdot y_1^N, u_1^{i-1} \oplus a_1^{i-1} | u_i \oplus a_i) \end{aligned} \quad (59)$$

Proposition 12: If a B-DMC W is symmetric, then W^N is also symmetric in the sense that

$$W^N(y_1^N | x_1^N \oplus a_1^N) = W^N(x_1^N \cdot y_1^N | a_1^N) \quad (57)$$

for all $x_1^N, a_1^N \in \mathcal{X}^N, y_1^N \in \mathcal{Y}^N$.

for all $u_1^N, a_1^N \in \mathcal{X}^N, y_1^N \in \mathcal{Y}^N, N = 2^n, n \geq 0, 1 \leq i \leq N$.

Proposition 12 and Proposition 13, Arikan, E. (2009)

The next section on encoding and decoding provided detailed descriptions of the algorithms. It focuses on the Successive Cancellation (SC) decoding algorithm, which exploits the recursive structure of the code and polarised nature of the channels. The encoding process involves mapping information bits to the selected information set of the polarised channels. SC decoding operates by sequentially decoding the information bits, leveraging the previously decoded bits to improve the reliability of subsequent decisions. The algorithm's complexity is relatively

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low at $O(\log(N))$, where N is the block length, making it practical for implementation. Theorem 3 is a critical result that links channel polarisation to the capacity-achieving property of polar codes as it states that for any given B-DMC and a fixed rate below the channel's symmetric capacity, the block error probability for polar coding under SC decoding can be made arbitrarily small by increasing the block length.

Theorem 3: For any given B-DMC W and fixed $R < I(W)$, block error probability for polar coding under successive cancellation decoding satisfies

$$P_e(N, R) = O\left(N^{-\frac{1}{4}}\right). \quad (15)$$

Theorem 3, Arikan, E. (2009)

Essentially implying that polar codes, with their specific encoding and decoding methods, can achieve the channel capacity of B-DMCs. Proposition 2 provided an upper bound on the block error probability for polar coding under SC decoding and the bound is expressed in terms of Bhattacharyya parameters of the channels in the information set, highlighting the influence of channel reliability on the overall decoding performance.

Proposition 2: For any B-DMC W and any choice of the parameters $(N, K, \mathcal{A})$

$$P_e(N, K, \mathcal{A}) \leq \sum_{i \in \mathcal{A}} Z\left(W_N^{(i)}\right). \quad (13)$$

Hence, for each $(N, K, \mathcal{A})$, there exists a frozen vector $u_{\mathcal{A}^c}$ such that

$$P_e(N, K, \mathcal{A}, u_{\mathcal{A}^c}) \leq \sum_{i \in \mathcal{A}} Z\left(W_N^{(i)}\right). \quad (14)$$

Proposition 2, Arikan, E. (2009)

Arikan also discussed the relationship between polar codes and existing coding techniques, specifically Reed-Muller (RM) codes, a well-known class of codes with recursive structures. He highlighted that polar codes share some similarities with RM codes, particularly in their recursive construction methods and the use of SC decoding. The concept of coset codes were also introduced and RM codes and polar codes were shown to be a specific instance of coset codes.

1) *G_N -Coset Codes:* We first describe a class of block codes that contain polar codes—the codes of main interest—as a special case. The block lengths N for this class are restricted to powers of two, $N = 2^n$ for some $n \geq 0$. For a given N , each code in the class is encoded in the same manner, namely

$$x_1^N = u_1^N G_N \quad (8)$$

where G_N is the generator matrix of order N , defined above. For $\mathcal{A}$ an arbitrary subset of $\{1, \dots, N\}$, we may write (8) as

$$x_1^N = u_{\mathcal{A}} G_N(\mathcal{A}) \oplus u_{\mathcal{A}^c} G_N(\mathcal{A}^c) \quad (9)$$

where $G_N(\mathcal{A})$ denotes the submatrix of G_N formed by the rows with indices in $\mathcal{A}$.

G_N -Coset Codes as defined by Arikan

However, a key distinction is that while RM codes select their information set based on the Hamming weight of codewords, polar codes utilise a channel-specific criterion, the Bhattacharyya parameter, to select the most reliable channels for data transmission, contributing to the superior performance of polar codes compared to RM codes. The paper also discussed the non-trivial generalisations of the polarisation scheme to non-binary input alphabets, where the basic ideas are extended to create capacity-achieving codes for more general channel models, which is highly relevant for practical applications where non-binary data needs to be transmitted -- a more general form of channel combining and splitting, where the channel input alphabet is q -ary, where $q > 2$, could be considered. Arikan

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also raised open questions regarding the precise speed at which channels polarise as understanding the asymptotic rate of polarisation could lead to improved design and analysis techniques for polar codes. It was also mentioned that the concept of channel polarisation has the potential to impact other areas of information theory. For example, it might find applications in source coding, cryptography, or even in the analysis of biological systems. Overall, Arikan's paper introduced a groundbreaking technique, channel polarisation, and as such, this paper has had a profound impact on the field of information theory.

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Advocate View

As mentioned, Arikan's paper made a significant and lasting contribution to the advancement of information theory. The paper's strengths will be highlighted in the following section.

The novel technique of channel polarisation was formally introduced in this paper, which allowed the creation of capacity-achieving codes for symmetric B-DMCs. Prior to this, constructing such codes remained an unsolved problem since Shannon's work. Hence, this paper significantly impacted the field and spurred new research avenues and eventually led to the development of the 5G network. Polar codes are also promising candidates for improving the reliability and efficiency of wireless communication systems, particularly in scenarios with challenging channel conditions. The principles of channel polarisation can also be adapted for designing error-correcting codes for data storage devices, enhancing data integrity. Not to mention, Arikan's work exhibits meticulous mathematical rigour as the paper established key theorems, developed complex proofs and introduced innovative concepts that underpin the foundation of channel polarisation and polar coding. This robust theoretical framework has made the paper a cornerstone of information theory with more than 5000 citations to its name.

Even while tackling intricate mathematical concepts, Arikan presents his ideas with exceptional clarity by employing visual aids in the form of binary tree representations of recursive channel transformations in order to enhance the understanding of the polarisation process. The detailed explanations of the encoding and decoding algorithms also makes it easier for a wider audience to implement polar codes.

Arikan also connected his novel techniques to existing coding methods, notably, the RM codes. By drawing parallels and emphasising key distinctions, he positioned polar codes within the broader scope of information theory. This comparative approach clarified the advantages and limitations of polar codes compared to other established methods, which provides a deeper clarity to readers.

Practical considerations were also addressed on top of the paper's strong theoretical basis, which brings a balanced view to the subject overall. Arikan discussed the complexity of the SC decoding algorithm and explored ways to generalise channel polarisation to manage non-binary input alphabets, recognising the importance of adapting the technique to real-world communication scenarios.

While the field of information theory is constantly evolving and future research might unveil possibly better coding techniques or discover the limitations to the applicability of channel polarisation, Arikan's contribution stands as a significant milestone that has fundamentally changed the approach to coding design. Its influence is likely to persist even as the field continues to progress.

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Critic View

While Arikan's paper is undoubtedly significant, a critical review should also consider its limitations. Arikan himself mentioned that channel polarisation relies on the asymptotic behaviour of the channel transformation, hence, at short lengths, the polarisation effect might not be pronounced enough, leading to suboptimal performance compared to other coding techniques. This limits the effectiveness of polar codes in applications where short codewords are essential. He also pointed out that the paper focuses primarily on symmetric B-DMCs and did briefly mention some potential extensions to handle non-binary input alphabets. Designing polar codes with specific parameters requires determining the "good" bit-channels, which can be computationally expensive, especially for larger block lengths. Although Arikan proposed a statistical method for approximating code construction, the complexity of finding optimal polar codes remains a challenge for practical implementations due to its high complexity.

The paper primarily focussed on the SC decoding algorithm. While SC decoding exhibits low complexity, it is inherently suboptimal compared to the maximum likelihood (ML) decoding. The reliance on SC decoding might limit the achievable performance of polar codes, especially in scenarios where computational resources are not as constrained. The performance of RM codes, the precursor to polar codes, under ML decoding has received little to no attention. This might imply that perhaps the performance of polar codes would be better if a different decoding algorithm was used. Not to mention, since SC decoding is a sequential decoding process, it might have a higher decoding latency compared to other algorithms and hence pose a problem in applications with strict delay requirements.

The construction of polar codes relies on knowledge of the channel statistics but in some scenarios, channel statistics might be unknown or time-varying, posing challenges for optimal code design. Channel estimation errors and time-varying channel conditions can also degrade the performance of polar codes. Since polar codes rely on the assumption of a known and stationary channel to identify the reliable and unreliable bit-channels during construction, when the channel state information is imperfect, the code might be forced to operate at a lower effective rate or be more prone to making errors. Although imperfect information is a general challenge in communication systems in general, the performance of polar codes could be particularly sensitive to inaccuracies since their construction and decoding processes rely heavily on precise knowledge of the channel.

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